AUTONOMOUS FIREFIGHTING UAVs

Multispectral Fire Detection, Localization and Geolocation Pipeline

Final Semester Report

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Repo: https://github.com/NinjaMonkey06/VIP_FireDetection

Abstract

This report documents the full design, implementation, and evaluation of a multispectral wildfire detection and localization pipeline for autonomous firefighting UAVs, developed over the course of the spring semester. The system processes paired RGB and thermal infrared (IR) frames from UAV-mounted multispectral sensors and produces fire presence decisions, spatial fire boundary contours, and structured coordinate data ready for handoff to a fire propagation model.

The project evolved through several major architectural stages: an initial evaluation of a pretrained RGB-only YOLO model, a strategic pivot to a custom multispectral mid-fusion classifier trained on the FLAME-2 dataset, an iterative localization pipeline progressing from MSER-based bounding box detection through to physics-informed fire boundary contour extraction, and finally a fully specified geolocation module design ready for implementation. The final classifier achieved 99.87% accuracy on the held-out FLAME-2 test set and about 84% sensitivity on out-of-distribution drone imagery. The localization pipeline produces geometrically accurate fire boundary polygons suitable for fire wavefront arc representation in the downstream propagation model.

1. Introduction

Wildfire detection and response represent a growing challenge in the context of climate change, with fires increasingly exceeding the capacity of conventional ground-based monitoring and suppression systems. Autonomous UAVs equipped with multispectral sensors offer a promising platform for real-time aerial fire monitoring, enabling rapid detection and localization of active fire fronts over large areas and in terrain inaccessible to ground crews.

This project addresses the perception and localization components of an autonomous firefighting UAV system. Given a continuous stream of paired RGB and thermal infrared frames captured during a UAV flight, the system must: (1) determine whether each frame contains an active fire, (2) spatially localize fire regions within fire-positive frames, and (3) produce structured coordinate outputs suitable for a fire propagation model and for handoff to a geolocation module that maps image-space detections to GPS world coordinates.

The system went through several major architectural stages over the course of the semester. This report documents each stage in full, including the rationale for each transition, to provide a complete record of the design evolution.

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2. Stage 1 — Pretrained RGB Detection (PyroNear YOLOv11)

2.1 Motivation and Model Selection

The first stage, carried over from the previous semester, focused on identifying an immediately available pretrained wildfire detection model to establish a baseline. The PyroNear YOLOv11 RGB model was identified as the most readily deployable option. RGB-Thermal (RGBT) models would theoretically offer improved discrimination between smoke and fire, but no ready-to-use wildfire-trained RGBT weights were publicly available. Implementing an RGBT model from scratch would have required constructing a custom labeled bounding-box dataset, a labor-intensive process not feasible within the project timeline.

2.2 Testing and Observations

The model was tested across three categories: frames containing active fire, frames containing only smoke, and clear frames with no fire or smoke, from a range of camera angles and altitudes. Performance on direct fire and clear scenes was reliable. Performance on smoke-only scenes degraded at low angles or low smoke density, attributed to the absence of thermal data. Bounding boxes frequently enclosed both fire and smoke together, and in scenes where smoke extended over large sky portions, the box covered the plume rather than the fire source, which is problematic for pixel-to-world coordinate conversion.

2.3 Decision to Transition

- Thermal infrared data encodes heat intensity directly and is unaffected by smoke opacity, making it intrinsically better suited for fire localization than RGB alone.
- The FLAME-2 dataset provides synchronized RGB and thermal frame pairs with frame-level labels, enabling a custom multispectral classifier without requiring bounding box annotations.

These observations motivated the transition to a custom multispectral pipeline.

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3. Stage 2 — Architectural Decision: Hybrid Pipeline

3.1 Literature Review

A review of Chen et al. (2022) established the conceptual foundation: thermal pixel intensity encodes heat emission and therefore flame location, enabling localization via classical image processing rather than supervised bounding box detection. This eliminates manual annotation at scale and is well-suited to edge deployment.

3.2 Final Architecture

```
Stage 1: RGB + Thermal Frame Classification
              ↓  (if fire detected)
Stage 2: Thermal Thresholding (scaled Otsu)
              ↓
Stage 3: Mask Dilation (merge nearby fire regions)
              ↓
Stage 4: Contour Extraction and Simplification
              ↓
Stage 5: Geolocation (designed, not yet implemented)
```

3.3 Why Not End-to-End YOLO?

Factor	YOLO Detection	Hybrid Pipeline
Manual bounding box labeling	Required	Not required
Compute cost	High (every frame)	Lower (clear frames skip localization)
Localization interpretability	Data-driven	Physics-driven, explainable
Real-time feasibility	Moderate	High
Scalability	Limited by annotation cost	Highly scalable

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4. Dataset — FLAME-2

4.1 Overview

Property	Value
Total frame pairs	53,451
RGB resolution	3840 × 2160 (4K)
Thermal resolution	640 × 512
Modality format	RGB (.jpg) + Thermal (.jpg)
Label classes	NN (no fire), YN (fire, no heavy smoke), YY (fire + heavy smoke)
Fire labeling criterion	Fire visible in RGB and/or thermal channel
Smoke labeling criterion	Smoke occupies ≥50% of RGB frame
Total dataset size	~126 GB

4.2 Sampling and Preprocessing

Regular interval sampling across labeled temporal segments was applied to 5,000 frame pairs to maximize scene diversity and reduce near-duplicate frames. RGB and thermal modalities were spatially aligned via center crop and a 6-pixel vertical offset correction. Both modalities were resized to 256×256. No ImageNet normalization was applied; the model trains on raw [0,1] pixel values from ToTensor() only, and this must be replicated exactly at inference time.

4.3 Dataset Split

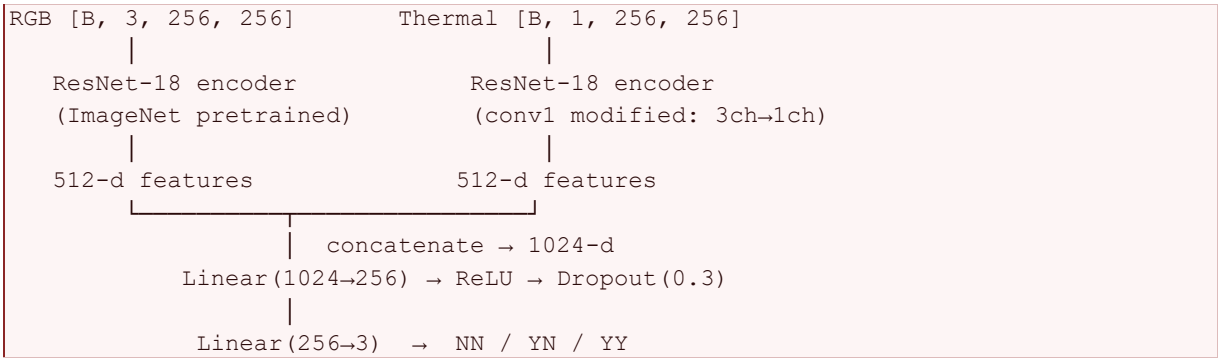
Split	Proportion	Approximate Size
Train	70%	~3,500 pairs
Validation	15%	~750 pairs
Test	15%	~750 pairs

Note: The split was made by frame rather than by video segment, so frames from the same flight segments appear in both train and test. The 99.87% test accuracy in Section 5.3 is therefore likely optimistic; the out-of-distribution results in Section 5.4 are the more reliable measure. A segment-level re-split is planned.

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5. Stage 3 — Multispectral Classifier (Phase 1)

5.1 Architecture: MidFusionNet



5.2 Training Configuration

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	16
Epochs	15
Loss function	Cross-Entropy
Dropout	0.3 (fusion head)

5.3 Results on FLAME-2 Test Set

	Predicted NN	Predicted YN	Predicted YY
Actual NN	255	0	0
Actual YN	0	248	1
Actual YY	0	0	246

Overall accuracy: 99.87%. Single misclassification: YN predicted as YY — fire correctly detected in both cases, only smoke classification differing.

5.4 Out-of-Distribution Testing

Batch	Content	Result
1	8 frames: 4 pos, 2 neg, 2 unlabeled	4/4 TP; 1/2 TN, 1 FP
2	23 negative frames	23/23 TN (100%), high confidence
3	116 negative frames (anomalous)	70/116 FP — thermal palette domain shift
4	23 negative frames (similar palette)	23/23 TN with high confidence
5	138 positive frames	116/138 TP (~84% sensitivity on OOD data)

Across clean batches: specificity **≥96%**, OOD sensitivity **~84%**. Batch 3 anomaly attributed to thermal palette domain shift; a different sensor's dynamic range mapping rendered ambient temperature as high-intensity pixels indistinguishable from fire in normalized pixel space.

5.5 Inference Script: infer.py

```
python infer.py --rgb_dir custom_test/rgb \
                --thermal_dir custom_test/thermal \
                --checkpoint fire_classifier.pth [--visualize]
```

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6. Stage 4 — Localization Pipeline Evolution (Phase 2)

6.1 v1 — MSER Bounding Boxes

The initial localization pipeline applied Otsu thresholding to produce a binary fire mask, MSER to detect stable high-intensity blobs, mask-MSER fusion to filter thermally unconfirmed regions, and IoU-based NMS to reduce overlapping boxes.

Observed Limitations

- Up to 44 bounding boxes per frame — MSER detected many small stable sub-regions within a single fire area.
- One frame produced zero detections — fire blob thermal gradient too soft for MSER stability threshold (delta=5).
- Containment nesting: small boxes inside large boxes survived NMS due to low IoU despite full spatial redundancy.

6.2 v2 — Tuned MSER with Containment Suppression

Parameter	v1	v2	Rationale
mser_delta	5	3	Catches softer thermal gradients; resolves zero-detection
mser_min_area	60 px	150 px	Filters noise fragments; reduces box count
mser_max_area	4000 px	6000 px	Captures large contiguous fire regions
nms_iou_threshold	0.3	0.2	More aggressive suppression
mask_overlap_ratio	0.40	0.35	Recovers soft thermal boundary detections
Containment suppression	None	0.85	Suppresses boxes ≥85% inside a larger box

Results: 2–15 boxes per frame. Zero-detection resolved. Containment nesting eliminated.

6.3 v3 — Transition to Contour Extraction

Supervisor feedback requested higher spatial precision and arc-shaped fire front representation. The fundamental limitation of bounding boxes was identified: a rectangle enclosing a crescent-shaped fire front discards all arc geometry. Version 3 replaced MSER, mask fusion, and NMS with fire boundary contour extraction directly from the Otsu mask using cv2.findContours() + cv2.approxPolyDP() simplification.

Observed Limitations of v3

- Nearby fire regions extracted as separate contours — should be a single front.
- Some small fire areas missed — pure Otsu threshold too high when fire is a small frame fraction.
- Moderate vertex density — arc fidelity could be improved.

6.4 v4 — Refined Contours (Final)

- **Threshold scaling (0.85):** multiplies Otsu threshold before applying it, expanding the fire mask to capture dimmer regions. Raw Otsu and applied values both logged per frame.
- **Mask dilation (kernel 15):** merges nearby fire regions before contour extraction. Regions within ~7px of each other merge into one contour — the physically correct representation for a connected fire front.
- **Denser vertices (epsilon 0.01):** lowered from 0.02, retaining more contour vertices for better arc coverage of the fire boundary.

6.5 Localization Evolution Summary

Version	Approach	Output	Key Limitation Addressed
v1	MSER + NMS	Up to 44 boxes/frame	Baseline
v2	MSER + NMS + containment suppression	2–15 boxes/frame	Box count, zero-detection, nesting
v3	Contour extraction (findContours + approxPolyDP)	Fire boundary polygons	Geometric accuracy for arc representation
v4 (final)	Contour + dilation + threshold scaling	Merged polygons, denser arcs	Region merging, missed fire, vertex density

6.6 Sample Output

inference_results.json — per frame


```
{
  "summary": {
    "total_frames": 60,
    "fire_detected": 40,
    "clear_frames": 20,
    "fire_rate": 0.6667
  },
  "results": [
    {
      "frame_id": "frame_001",
      "filename": "frame_001.jpg",
      "classifier_prediction": "YN",
      "fire_detected": true,
      "phase2_localization": true,
      "probabilities": { "NN": 0.02, "YN": 0.91, "YY": 0.07 }
    }
  ]
}
```


localization_results.json — per fire-positive frame


```
{
  "summary": {
    "localization_run_on": 40,
    "frames_with_regions": 38,
    "fire_detected_no_regions": 2,
    "parameters": {
      "threshold_scale": 0.85,
      "merge_kernel_size": 15,
      "epsilon_factor": 0.01
    }
  },
  "results": [
    {
      "frame_id": "frame_001",
      "fire_detected": true,
```

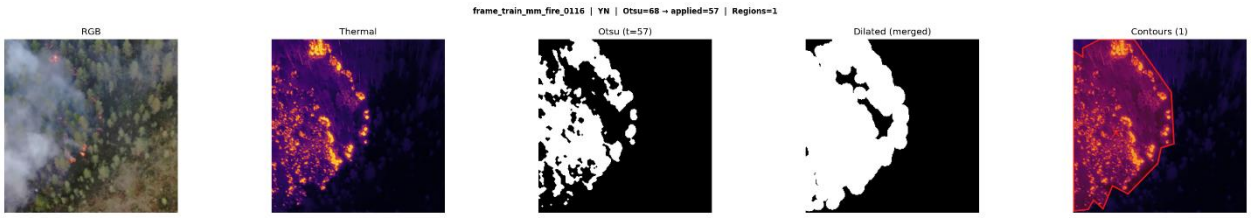
```
      "localization_run": true,
      "otsu_threshold": 187,
      "applied_threshold": 158,
      "n_raw_contours": 3,
      "fire_regions": [
        {
          "pixel_256": [{"x":112,"y":88},{"x":134,"y":91},{"x":145,"y":103},
"..."],
          "normalized": [{"x":0.4375,"y":0.3437},{"x":0.5234,"y":0.3554}, "..."],
          "area_px": 1823.5,
          "n_points": 14,
          "original_wh": {"width": 640, "height": 512},
          "gps_polygon": null,
          "fire_front_arc": null
        }
      ]
    }
  ]
}
```


6.7 Sample Visualization Outputs

The figures below show representative outputs from the localization pipeline at two stages of development. Each frame is rendered as a multi-panel figure produced by the --visualize flag.



Bounding Box Output (v2): RGB | Thermal | Otsu Mask | Detections with bounding boxes overlaid



Contour Output (v4 final): RGB | Thermal | Otsu Mask | Dilated Mask | Fire boundary contours overlaid

7. Stage 5 — Geolocation Module Design (Phase 3)

The geolocation module converts normalized image-space fire contour points from Phase 2 into GPS world coordinates. It has been fully designed but not implemented, and committed as a pseudocode stub (geolocate.py), pending a live telemetry log for implementation and validation.

7.1 Hardware Specification

Component	Specification
Thermal camera	FLIR HADRON 640R+, 13.6mm focal length, 60Hz
Altitude	Barometric + GPS fusion
Gimbal	Real-time pitch/roll/yaw angle telemetry
Telemetry format	MAVLink v2 over serial

7.2 Camera Intrinsics

Parameter	Value	Derivation
Sensor resolution	640 × 512 px	Datasheet
Pixel pitch	12 µm	Datasheet
Focal length	13.6 mm	Supervisor confirmation
Focal length (pixels)	1133.3 px	= 13.6 / 0.012
Horizontal FoV	~50.4°	= 2 × arctan(640 / (2 × 1133.3))
Scaled fx (256px)	453.3 px	= 1133.3 × (256/640)
Scaled fy (256px)	566.7 px	= 1133.3 × (256/512)

7.3 Back-Projection Method

- **Step 1:** pixel → camera ray — `ray_cam = K⁻¹ × [u, v, 1]ᵀ`
- **Step 2:** rotate to world — `ray_world = R × ray_cam` (R from gimbal Euler angles)
- **Step 3:** ground plane intersection — `t = altitude_m / ray_world[2]` (flat-earth assumption)
- **Step 4:** metre offset → GPS delta via spherical earth approximation

7.4 Fire Wavefront Arc and Output Formats

Each GPS polygon's leading fire edge is extracted using wind direction alignment (edges whose outward normals face into the wind vector). Without wind data, the full polygon is output. Three output formats are supported: JSON (pipeline-internal), GeoJSON (GIS tools), and CSV (propagation model input).

8. Key Design Decisions

Decision	Justification
Hybrid DL + classical CV	No localization cost on clear frames; physics-grounded, explainable localization; no bounding box annotation required.
Mid-fusion over early/late fusion	Allows modality-specific feature development before cross-modal integration.
Otsu over fixed threshold	Adapts to per-frame intensity histograms; robust across sensors without per-dataset calibration.
Contours over bounding boxes	Bounding boxes discard fire boundary geometry. Contours directly trace the fire front arc required by the propagation model.
Mask dilation for merging	Simpler and more efficient than post-hoc contour merging. Physically correct: nearby fire patches form one front.
Threshold scaling (0.85)	Expands mask for small fire regions where Otsu overestimates threshold, while retaining per-frame adaptivity.
Dual coordinate output	Resolution-agnostic. Normalized coords are platform-independent and directly consumable by geolocation.
Visualization as CLI flag	Enabled for development; disabled for deployment by omitting --visualize. No code changes between modes.

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9. Known Limitations

- **Thermal palette domain shift:** the classifier was trained on a single thermal sensor platform. Different cameras apply different color palette mappings, which produced systematic false positives in Batch 3 testing. Mitigation: cross-sensor normalization or multi-platform fine-tuning.
- **Distance and angle effects on thermal signature:** at long range or oblique angles, the thermal signature of tree canopies can resemble that of a fire source, leading to false positive detections in the localization stage. At close range and near-nadir angles this confusion does not occur. This should be considered when setting UAV flight parameters for operational deployment.
- **Fixed processing resolution:** both modalities are resized to 256×256. Small or distant fire regions may lose spatial detail. A multiscale approach could improve small-source detection.
- **No temporal modeling:** frames are processed independently. Temporal correlation across a UAV video stream is unused. Frame differencing or recurrent feature aggregation could improve consistency.
- **Flat-earth geolocation assumption:** geolocate.py assumes flat terrain at the drone's fused altitude. Mountainous terrain requires DEM intersection for accurate GPS projection.
- **Geolocation not yet implemented:** geolocate.py is a fully specified stub requiring a MAVLink v2 telemetry log, pymavlink, and pyproj to activate.
- **False positives on extreme thermal palettes:** atypically high global thermal intensity frames are classified as fire with high confidence. Fine-tuning with hard negatives from these conditions is recommended if observed in deployment.

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10. Future Work

- **Implement geolocate.py:** all math and interfaces fully specified. Requires MAVLink v2 telemetry log, pymavlink, pyproj.
- **End-to-end wrapper:** pipeline.py combining infer.py → localize.py → geolocate.py in one deployment command.
- **Wind direction integration:** activate wind-guided wavefront arc extraction via onboard sensor or weather API.
- **Multi-drone coordination:** distribute GPS fire coordinates across a UAV swarm for coordinated suppression planning.
- **Cross-sensor normalization:** normalize thermal images from different platforms to a common intensity scale.
- **Temporal modeling:** incorporate frame-to-frame context to improve consistency and reduce false positives.
- **YOLO comparison study:** benchmark classical CV localization against a trained YOLO detector on the same OOD test frames.
- **DEM-based geolocation:** replace the flat-earth assumption with DEM intersection for mountainous terrain.

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11. Conclusion

This project developed a complete multispectral fire detection, localization, and geolocation pipeline for autonomous firefighting UAVs across one semester, progressing through five major design stages: pretrained RGB-only YOLO evaluation, architectural pivot to a hybrid multispectral pipeline, custom classifier training on FLAME-2, iterative localization development from MSER bounding boxes to refined fire boundary contour extraction, and full geolocation module design.

The final classifier achieved 99.87% accuracy on FLAME-2 and about 84% sensitivity on out-of-distribution drone imagery. The localization pipeline evolved from up to 44 redundant bounding boxes per frame to geometrically accurate merged fire boundary polygons with dense vertex representation. The geolocation module is fully specified and ready for implementation. The pipeline is modular, documented, and committed to a public GitHub repository, providing a solid foundation for the geolocation, multi-drone coordination, and propagation model integration work that follows.

Team



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